



LECTURE 6

SPATIOTEMPORAL REPRESENTATIONS AND NEURAL FIELDS

Geospatial Representation Learning

PRESS – OR SPACE. →

© Marc Rußwurm

Licensed under [Creative Commons Attribution-NonCommercial 4.0 International \(CC BY-NC 4.0\)](#).

You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.

Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

Learning Outcomes

Lecture

- Explain how temporal dynamics can be represented in geospatial machine learning models.
- Describe neural fields and implicit neural representations for continuous spatial and spatiotemporal data.
- Compare grid-based, sequence-based, graph-based, and coordinate-based representations of Earth system data.
- Assess neural fields for interpolation, compression, forecasting, and multimodal integration.

Lab

- Prepare a small spatiotemporal geospatial dataset for representation learning.
- Implement or use a simple coordinate-based neural representation.
- Evaluate interpolation or reconstruction quality across space and/or time.
- Analyze continuous neural representations compared to discrete raster-based approaches.



PRACTICAL 6

LEARNING A SPATIOTEMPORAL REPRESENTATION

Geospatial Representation Learning

PRESS – OR SPACE. →

© Marc Rußwurm

Licensed under [Creative Commons Attribution-NonCommercial 4.0 International \(CC BY-NC 4.0\)](#).

You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.

Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.